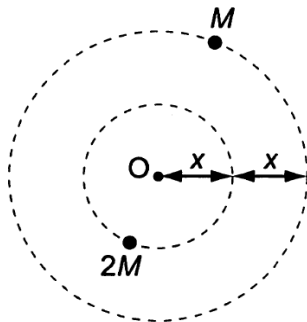


12

Two stars of mass M and $2M$, a distance $3x$ apart, rotate in circles about their common centre of mass O .



What is the angular speed of the star of mass $2M$?

A

$$\frac{1}{3} \sqrt{\frac{GM}{x^3}}$$

B

$$\frac{1}{3} \sqrt{\frac{2GM}{x^3}}$$

C

$$\frac{1}{2} \sqrt{\frac{GM}{x^3}}$$

D

$$\sqrt{\frac{GM}{x^3}}$$

L2

Answer: A

[Turn over]

For each star's circular motion, the gravitational force exerted by the other star provides the centripetal force required.

Considering star of mass $2M$,

$$F_G = F_c$$

$$\frac{GM(2M)}{(3x)^2} = (2M)x\omega^2$$

$$\omega^2 = \frac{GM}{9x^3}$$

$$\omega = \sqrt{\frac{GM}{9x^3}} = \frac{1}{3} \sqrt{\frac{GM}{x^3}}$$